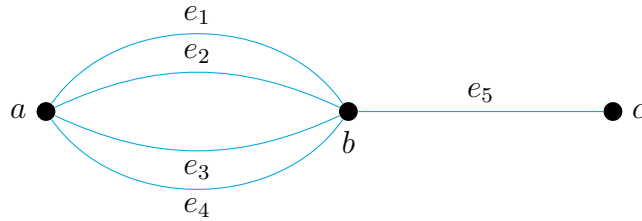


10.1.5

Consider the following graph:



- How many paths are there from a to c?
- How many trails are there from a to c?
- How many walks are there from a to c?

- How many paths are there from a to c?

Path = no repeated vertices

ae_1be_5c
 ae_2be_5c
 ae_3be_5c
 ae_4be_5c

Solution: a. There're 4 paths from a to c.

- How many trails are there from a to c?

Trail = no repeated edges

Since all trails must end with be_5c , and there're four edges, where none may be repeated, then the possible trails of the form $a \rightarrow b \rightarrow a \rightarrow b \rightarrow c$ is:

$$P(4, 3) = \frac{4!}{(4-3)!} = 4 \times 3 \times 2 = \boxed{24}$$

And then for the valid form $a \rightarrow b \rightarrow c = 4$. Therefore:

$$24 + 4 = \boxed{28}$$

Solution: b. There're 28 trails from a to c.

c. How many walks are there from a to c?

Walk = valid sequence of adjacent vertices/edges

Since repetition of vertices and edges are now allowed, we can conceptually conclude that we can walk between a and b arbitrarily many times before finally taking the final walk from b to c.

Solution: c. There are infinitely many walks from a to c.